

automation (EDA) tools. Last, VLSI fabrication/prototyping is expensive and there is no commercial VLSI semiconductor manufacturing facility (foundry) in India.

K. Srinivasa Raju, CEO, Unistring Tech Solutions Pvt Ltd, informs that there are very few openings for jobs in analogue designing in the VLSI industry. He says, "The expectations in terms of the know-how of analogue complementary metal oxide semiconductor (CMOS) designs/issues are very high, which makes it difficult to get into analogue based VLSI companies. Most of the companies prefer to take M.Techs from only reputed institutes such as NITs/IITs."

However, Sen believes, as India expands its electronics system design and manufacturing (ESDM) capabilities, the number of job opportunities is expected to grow in the VLSI design area.

Mostly verification based jobs in India

Jobs in this industry are broadly distributed in four areas: FPGA based embedded system design, embedded based small FPGA application development, application specific integrated circuits (ASIC) based designs and VLSI verification for FPGAs/ASICs/embedded based designs.

"Most of the jobs in the VLSI industry in India are verification based, particularly for ASIC designs," informs Raju. To get into such companies, one must be good in SystemVerilog and unified verification methodology (UVM) or open verification methodology (OVM)." He adds, "Most of these verification based companies prefer M.Tech graduates specialised in VLSI design."

Good opportunities for fresh graduates

There are many career opportunities in the VLSI industry today, especially at the entry-level roles targeted at fresh engineering graduates. "One needs to understand that VLSI, as seen from textbooks, is not

Why students from IITs/NITs are preferred for design

Most of the companies, including multinational corporations (MNCs), turn to IITs/NITs for recruitment of graduates for VLSI design mainly because these institutes have the latest and sophisticated computer aided design (CAD) VLSI tools and FPGA (field programmable gate array) boards with experienced professors. Also, their training methodology and lab exercises on FPGA implementation in VHDL [VHSIC (very high speed integrated circuit) hardware description language]/Verilog HDL are not limited to their academic course syllabus. Raju notes, "These students are trained on latest emerging trends in the VLSI domain. Professors extract a lot from students by giving real-time problems and assignments. Most of the training is hands-on and more practical, which is not the usual practice in the regular engineering colleges."

Skills required at a glance

- ▶ Strong fundamentals in digital electronics (also referred to as digital logic design/switching theory and logic design) and digital system design.
- ▶ Fundamental knowledge of digital circuit design with exposure to design using hardware description languages (such as VHDL and Verilog), functional simulation and logic synthesis.
- ▶ For VLSI verification, knowledge of SystemVerilog and verification methodology is a must.
- ▶ Basic understanding of functional verification and proficiency in scripting using Perl/tool command language (TCL).
- ▶ Career prospects can be improved by building skills in digital CMOS IC design, physical verification flow, analogue circuit modelling and layout analysis.
- ▶ Knowledge of VLSI design flows and proficiency in industry standard EDA tools for design of ASICs, FPGAs and system on chip (SoC) are also highly valuable.
- ▶ Strong fundamentals in discrete op-amp based analogue circuit design and analysis, digital logic design and analysis, and microprocessors/microcontrollers programming.

just limited to that. Skills such as digital domain and VHDL/Verilog HDL languages can be acquired that are used in FPGA based embedded system industry and also in ASIC companies," says Raju.

Excellent fundamentals in digital electronics, topped with knowledge in either VHDL or Verilog HDL, can easily get an engineer a job in FPGA based companies. Especially for an M.Tech graduate in VLSI, at entry level it is highly possible to get an opportunity to work at the design level in the VLSI industry.

Embedded system design companies that develop FPGA based embedded applications also have openings for fresh graduates. Raju notes, "At the entry level, as fresh undergraduates (B.Tech/B.E.), it is comparatively easier to get into FPGA based embedded system companies than ASIC-VHDL/Verilog HDL based companies."

This field offers fresh engineering graduates opportunities in several stages of the VLSI chip design pro-

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cess too. "The biggest opportunities continue to be in the front-end register transfer level (RTL) design and verification with growing opportunities in logic synthesis and timing analysis, design for testability, physical design and verification, analogue and mixed signal CMOS IC design, CAD tools development, and hardware verification and validation," according to Mysore.

Fresh graduates also have opportunities in field application engineering, technical support, and marketing and sales.